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ABSTRACT

The **Income Tax Management and Calculation System** is a web-based application developed using PHP, HTML, CSS, JavaScript, and Python for automation. It is designed to simplify and streamline the process of tax filing, revenue tracking, and professional communication for taxpayers. The system enables users to manage their tax-related information efficiently, upload financial documents, assign tax professionals, and receive real-time notifications.

An integrated automation module built with Python enhances accuracy and speed in calculations, document processing, and tax estimation based on the latest tax regimes. The platform also includes an interactive tax calculator that dynamically computes estimated taxes using the Indian tax structure.

The system is designed to simplify tax computation, document handling, and professional communication through a secure, automated workflow

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Chapter 1: Introduction

1.1 **Overview of the System**

The Income Tax Calculation and Management System is a web-based application developed using PHP, HTML, CSS, JavaScript, and Python for automation. The system is designed to simplify the process of managing income tax-related activities such as calculating taxes, maintaining taxpayer details, assigning tax professionals, and securely uploading necessary financial documents.In the current digital era, manual tax filing processes are time-consuming and prone to human errors. This system automates these operations by providing taxpayers with a unified digital platform to manage their financial records and tax estimations in accordance with the latest Indian tax regime

1.2 Problem Statement

In traditional income tax filing and management systems, most processes are manual, complex, and repetitive. Maintaining and calculating their annual income and deductions, Identifying applicable tax slabs and computing payable tax, Tracking communications between taxpayers and professionals, Managing document uploads and secure data handling These challenges can result in delayed tax submissions, inaccurate filings, and lack of transparency. Therefore, there is a need for a **digital solution** that integrates automation, accuracy, and user-friendliness to manage all tax-related tasks efficiently

1.3 Objectives

The main objectives of the **Income Tax Management System** project are:

- To develop a web-based platform that enables taxpayers to manage their income, deductions, and taxes digitally.

- To automate the tax calculation process using Python, ensuring compliance with the **new Indian tax regime**.

- To provide a secure document upload and storage system for financial and identification files.

- To facilitate communication between taxpayers and tax professionals through notifications and comments.

- To implement a secure login and registration system for users.

- To improve accuracy and reduce manual errors in tax computation and data entry.

- To enhance transparency, accessibility, and ease of use for all users through a responsive and intuitive user interface.

1.4 Organization of the Mini-Project Report

This report is organized as follows:

- Chapter 1 Describes the overview, objectives, and motivation behind the project.

- Chapter 2 Discusses existing tax management systems, their limitations, and how this project provides improvements.

- Chapter 3 Outlines the hardware and software requirements necessary for implementation.

- Chapter 4 explains the system design and architecture.

- Chapter 5 Presents the architectural design, data flow, and use case diagrams of the system.

- Chapter 6 Explains the coding, database connectivity, automation features, and web development modules..

- Chapter 7 Showcases the project output, user interface, and discussion of the results obtained.

- Finally, concludes the report by summarizing project outcomes, practical applications, and possible future improvements.

Chapter 2: Literature Survey

2.1 Existing Solutions

The management of income tax in India has largely evolved with the introduction of digital systems like the Income Tax e-Filing Portal, TRACES (TDS Reconciliation Analysis and Correction Enabling System), and several third-party tools such as Clear Tax, Quicko and TaxSpanner.

While these existing solutions have improved accessibility, most of them are either too complex for first-time users or require advanced financial knowledge to operate effectively. They often lack personalized automation and dynamic dashboards that simplify understanding of tax structures. Many of these platforms are built for large-scale government or enterprise use, which makes them less suitable for small businesses, students, or individuals with minimal tax knowledge.

2.2 Gaps Identified

- **Lack of Simplicity and Accessibility:** Modern taxpaying system is not beginner friendly.
- **Limited Automation in Calculation:** Many existing systems rely on static calculators without automation.
- **Minimal Integration with User Roles:** Present system cannot differentiate the taxpayer and professional modules.
- **Limited Customization for Indian Tax Regimes:** Many global or imported systems do not fully accommodate the unique Indian tax slab structure, especially when switching between old and new regimes
- **Data Security and Privacy Concerns:** Some platforms lack robust data handling mechanisms for sensitive tax documents. The proposed system ensures secure file uploads and storage using role-based access and session handling in PHP**.**

- Tao, Y., Zhang, Y., & Wang, X. (2022). Development of Personal Income Tax Management System Based on Data Mining Technology. Proceedings of the International Conference on Big Data and Intelligent Education (ICBDIE). Atlantis Press.
- Karpagavalli, V. (2017). E-Filing of Income Tax Returns. Indian Journal of Computer Science.
- Raut, S. H., & Shinde, P. R. (2021). Evaluation of Individual Perception Towards Filing Electronic Income Tax Return. Journal of Asian Research
- Dey, S. K. (2023). Taxpayers' Perceptions Towards Income Tax Administration: A Study from Indian Perspectives. Journal of Tax Reform, Vol. 9 (3), 300-316

Chapter 3: Software Requirement Specification

3.1 Hardware Requirements

Table 1: Hardware Requirements

Component	**Specification / Requirement**	**Description / Purpose**
Processor (CPU)	Intel i5 / AMD Ryzen 5 or higher	To handle backend processing and tax automation operations
RAM	Minimum 4 GB	Required for smooth running of AI model and web server
Storage	Minimum 100 GB HDD / SSD	To store financial documents, reports, tax records.
Graphics Card (GPU)	NVIDIA GTX 1050 / AMD equivalent (Optional)	No GPU needed; system uses simple Python automation
Internet Connection	Stable broadband (≥ 10 Mbps)	For API requests, dataset download, and web application

3.2 Software Requirements

Table 2: Software Requirements

Software Component	Version / Tool	Description / Purpose
Operating System	Windows 10/11	System that allows to run the task in an efficient way.
Front end: HTML, CSS, JavaScript	Latest stable	Frontend tools for creating responsive user interface.
Backend Technologies: PHP 8.0+	Latest stable	Web Frame work for handling the request sent by the front end.
Database	Latest	Store the credentials
Server environment	Latest stable	Xampp (MySQL,Apache).
IDE/Editor	Latest stable	Visual Studio Code for composing the code.
Web Browser	Chrome, Firefox, Edge	For accessing and testing the web application interface.

Chapter 4: System Design

4.1System Architecture

1. The **Admin** will manage the taxprofessional, taxpayer, taxauthority and their records.
2. The **PHP** backend implements the logic like: calculation of the tax, notifying, authority access, managing the records and other operations.
3. The **Automation** in the python will help to automate all the things and helps in tax calculation and analysis.
4. The **database(MySQL)** will help to gather all the credentials and help to retrieve them for the specific user.
5. The backend, frontend and automation will work coordinate for smooth work flow.

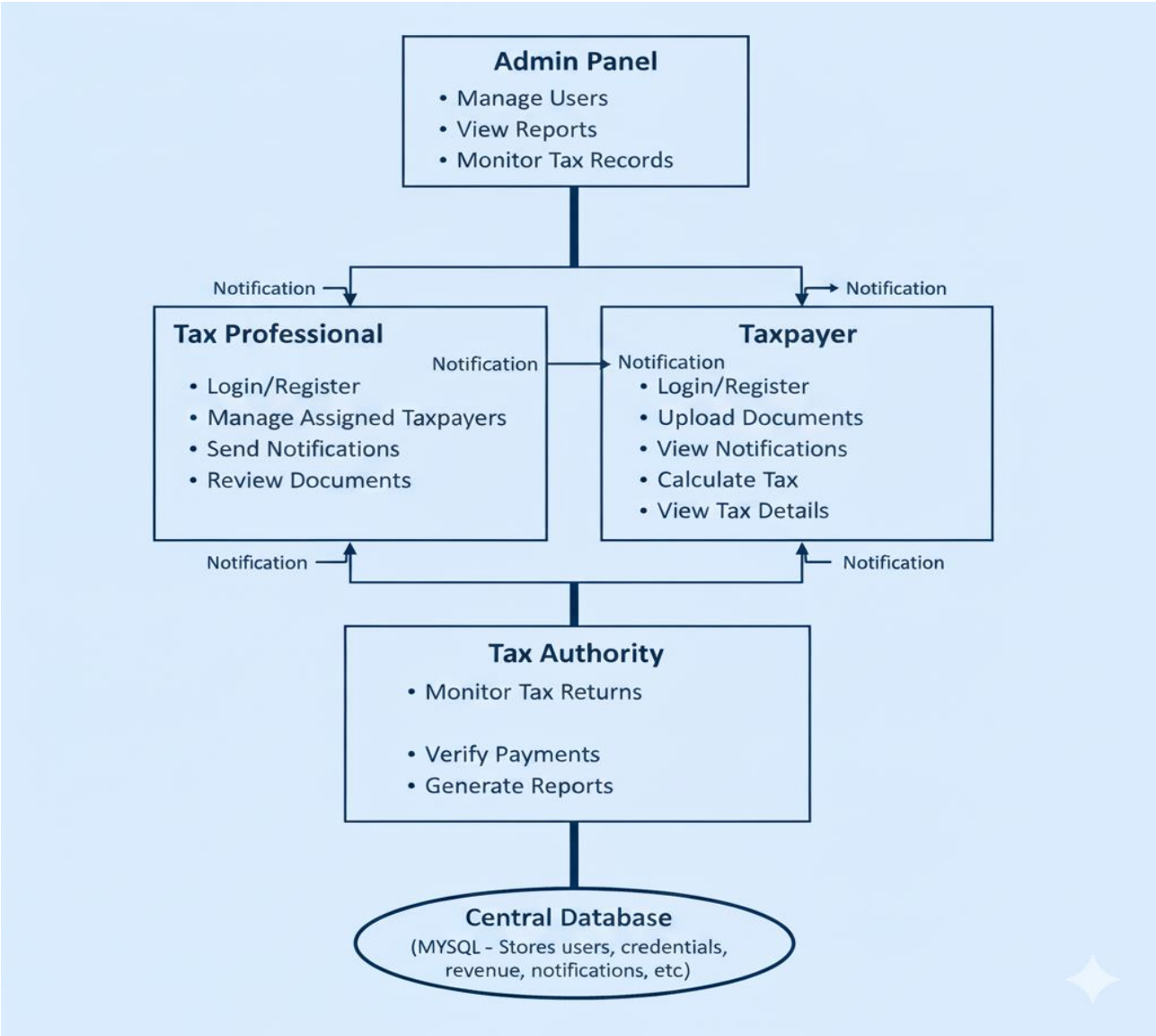

Fig 4.1 – System Architecture Diagram – ITMS

4.2 Use Cases

This use case diagram shows the interaction between users and the system functionalities.

- **Taxpayer:** Can log in, upload documents, add revenue details, calculate estimated tax, and view notifications
- **Tax Professional:** Manages assigned taxpayers, reviews their uploaded documents, and sends notifications/comments.
- **Admin:** Controls all system-level activities such as adding users, assigning roles, and generating reports.
- **Tax Authority:** Verifies tax submissions, validates compliance, and generates summary reports for audit.

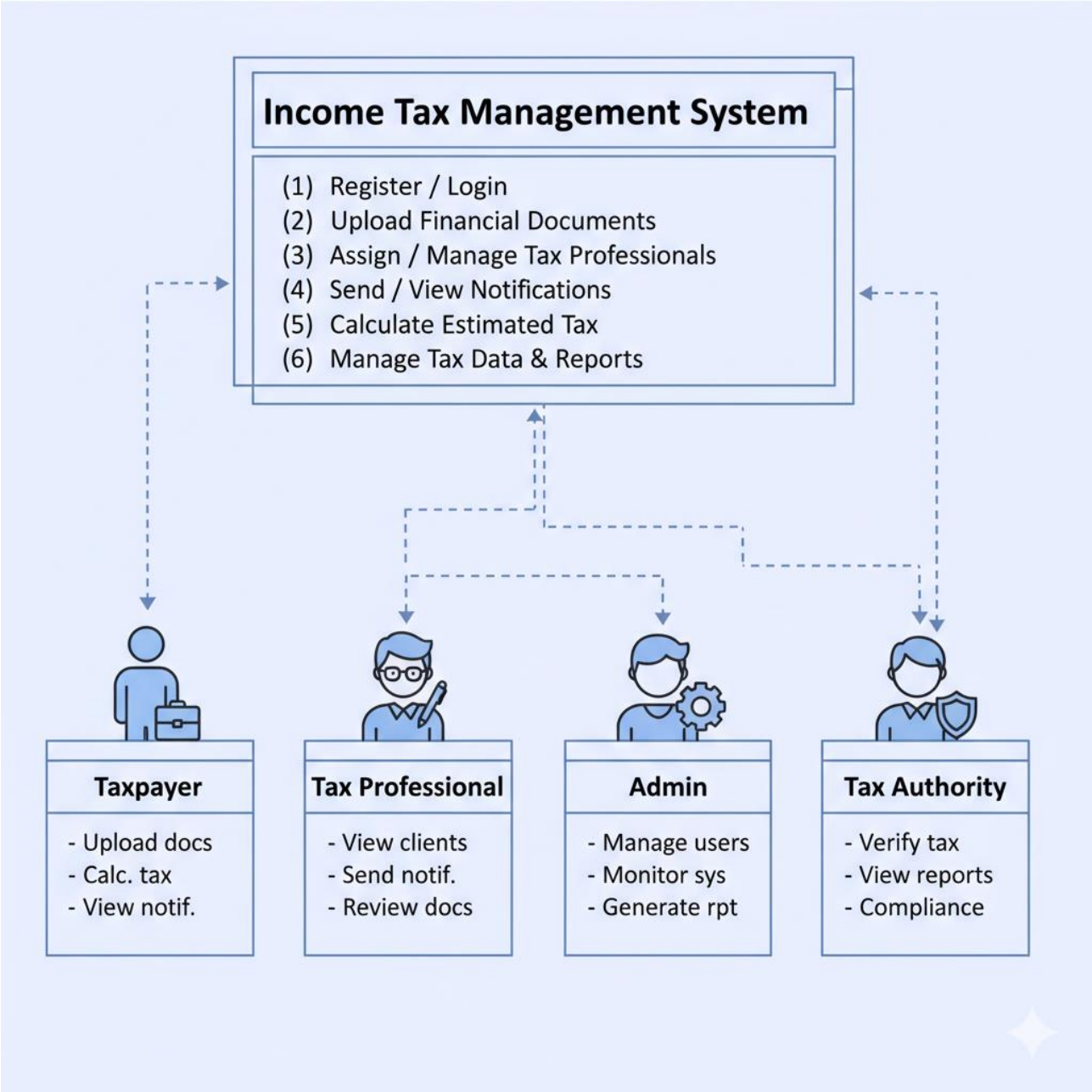

Fig 4.2 – User Case Diagram - ITMS

4.3 Data Flow Diagram

- **Data Input:** Taxpayer enters financial and income details.
- **Processing:** PHP and Python handle authentication, tax computation, and notification generation.
- **Storage:** All data (credentials, notifications, reports) is securely stored in a MySQL database.
- **Output:** Calculated tax, reports, and updates are displayed to the respective user roles.
- **Admin & Authority:** Access system-wide reports and ensure data integrity.

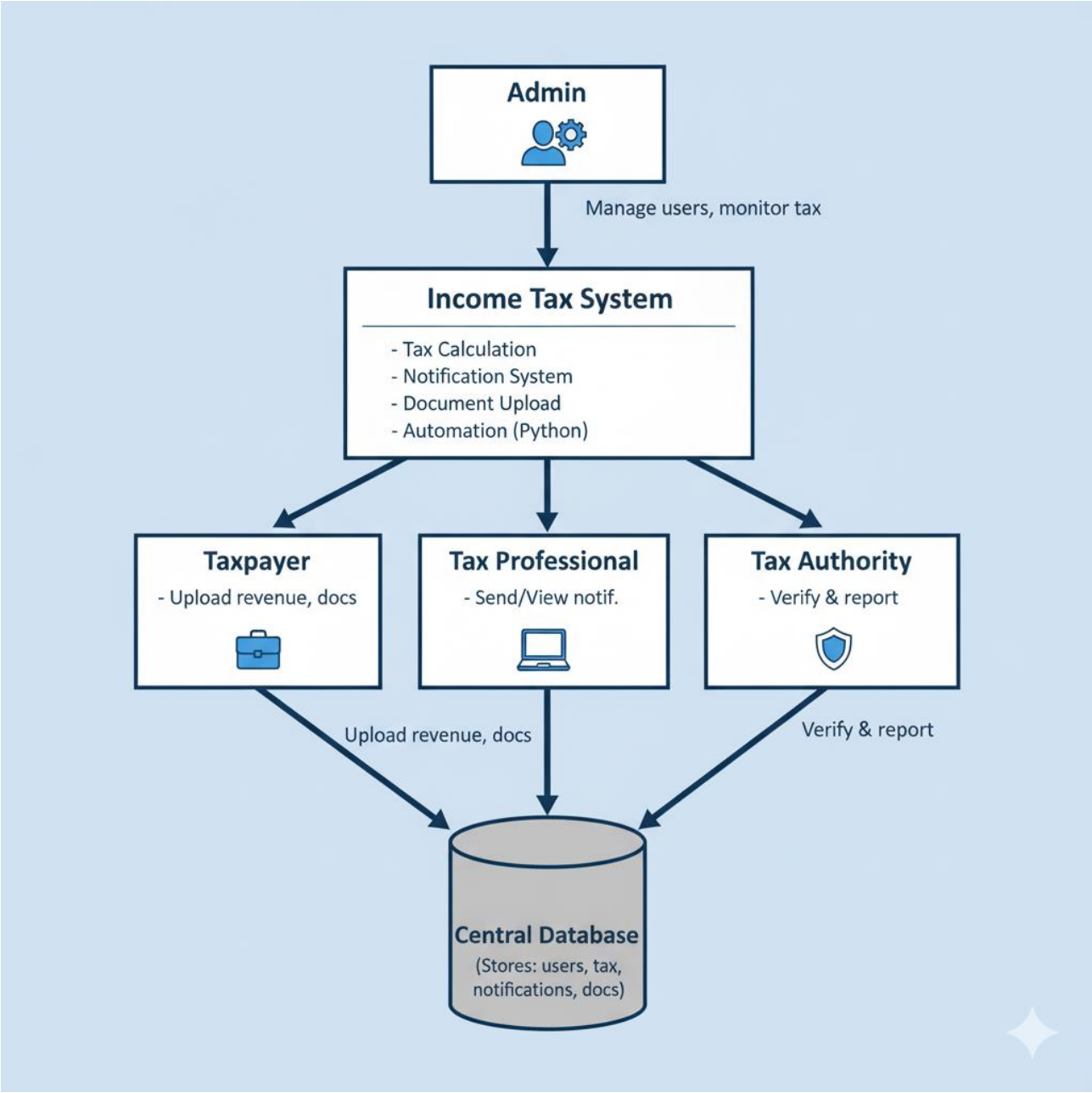

Fig 4.3 –Data Flow Diagram - ITMS

Chapter 5: Implementation

5.1Database Implementation

The **Income Tax Management System (ITMS)** database is designed by the relational model it ensures the normalization and efficient data integrity. It connects the **Admin, Taxpayer, Tax Professional, Tax Authority** using the primary and foreign keys. **MySQL** will handle the database and **PHP** will handle the dynamic data interaction.

Key Database Tables and Relationships:

1. **User Table**
   - Stores basic login credentials and personal details of every system user, including Admins, Taxpayers, Tax Professionals, and Tax Authorities.
     Fields:
     UserID (PK), Username, Password, Role, Email, Phone
2. **Taxpayer Table**
   - Maintains taxpayer-specific details and links each taxpayer to a user account and their assigned tax professional.
     Fields:
     TaxpayerID (PK), UserID (FK), Name, Email, Phone, TaxProfessionalID (FK)
3. **TaxProfessional Table**
   - Contains details of professionals managing taxpayer portfolios.
     Fields:
     ProfessionalID (PK), UserID (FK), Name, Email, Phone
4. **TaxAuthority Table**
   - Records officials from the tax authority who oversee overall tax operations and verify tax returns.
     Fields:
     AuthorityID (PK), UserID (FK), Designation, Department
5. **Admin Table**
   - Represents the system administrator responsible for managing all users, assigning roles, and monitoring system performance.
     Fields:
     AdminID (PK), UserID (FK), Name, Email
6. **Notification Table**
   - Enables communication between tax professionals and their assigned taxpayers. Each notification record is linked to a taxpayer.
     Fields:
     NotificationID (PK), UserID (FK), Message, Type, Timestamp
7. **TaxRevenue Table**
   - Tracks taxpayer income, deductions, and computed tax values.
     Fields:
     RevenueID (PK), UserID (FK), Amount, Date, TaxableIncome, TaxDue
8. **Document Table**
   - Stores file upload metadata for taxpayer-submitted financial documents.
     Fields:
     DocumentID (PK), UserID (FK), FileName, UploadDate, FilePath

5.2 Web Application Development

The ITMS application is developed using **PHP, HTML, CSS, and JavaScript** with a responsive frontend and secure backend.

Module Description	
Admin Module	Manages all users, assigns roles, and monitors tax records.
Taxpayer Module	Allows users to upload documents, add revenue, view tax calculations, and receive notifications.
Tax Professional Module	Manages assigned taxpayers, sends comments, and monitors document uploads.
Tax Authority Module	Oversees taxpayer compliance, verifies tax returns, and generates audit reports.

Figure 5.2.1: Home Page of ITMS Web Application

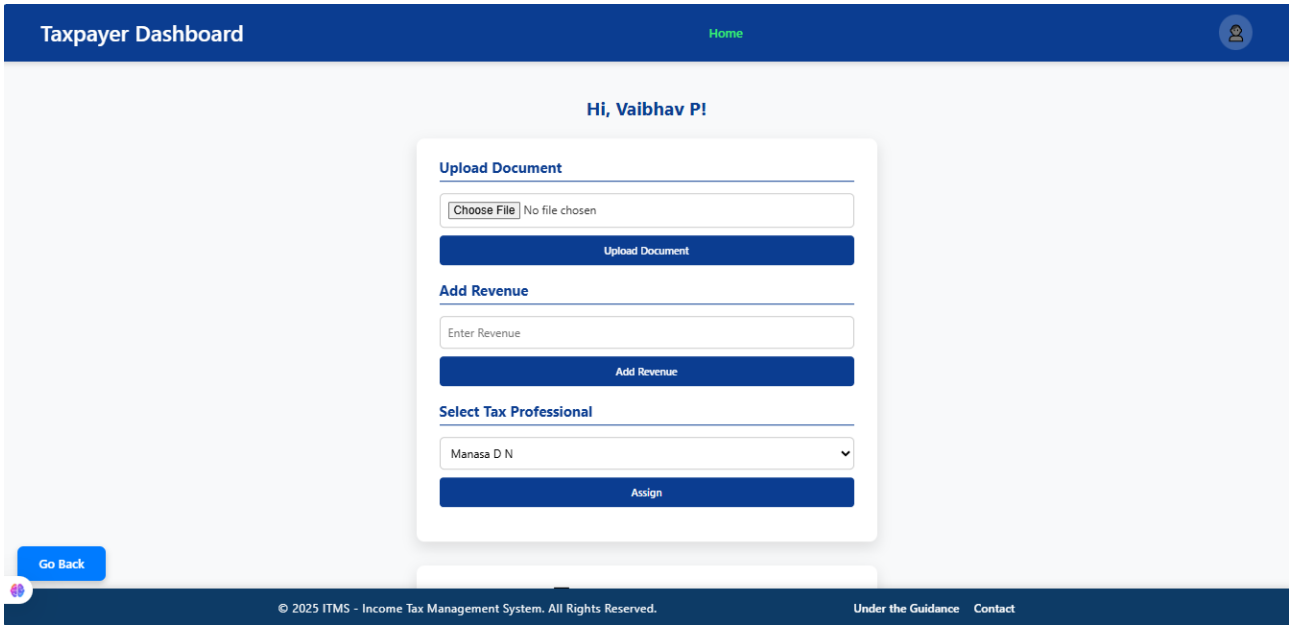

Figure 5.2.2: Tax Payer Home Page

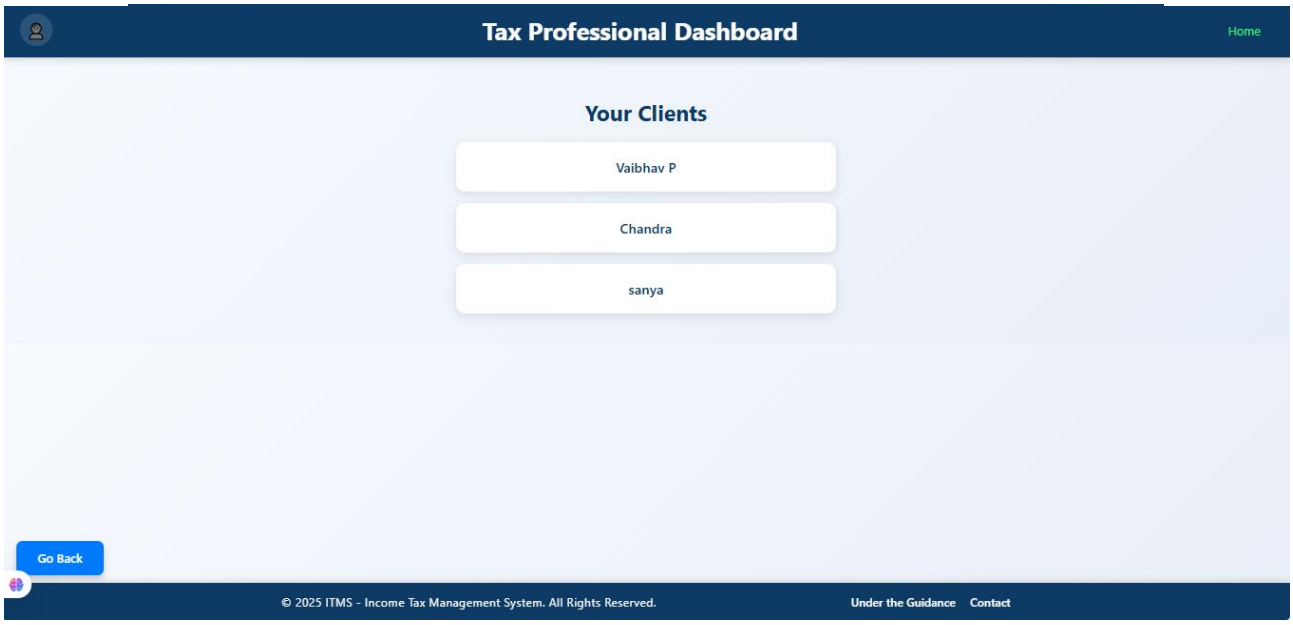

Figure 5.2.3: TaxPayer Dashboard

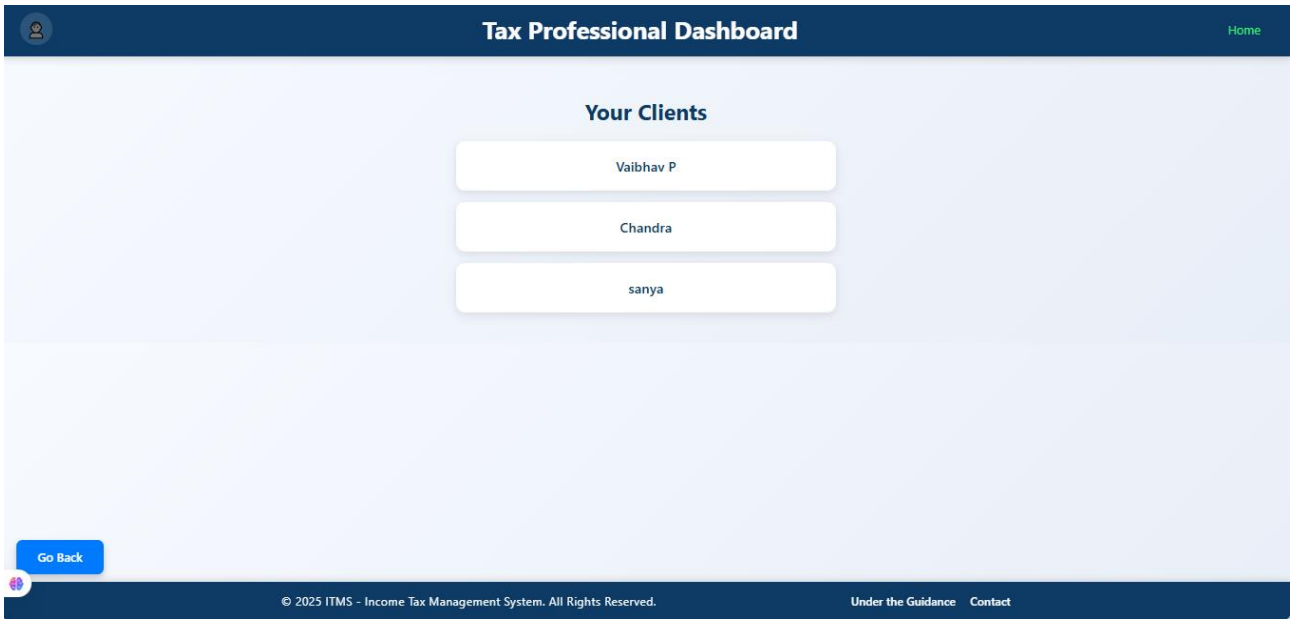

Figure 5.2.4:TaxProfessional Dashboard

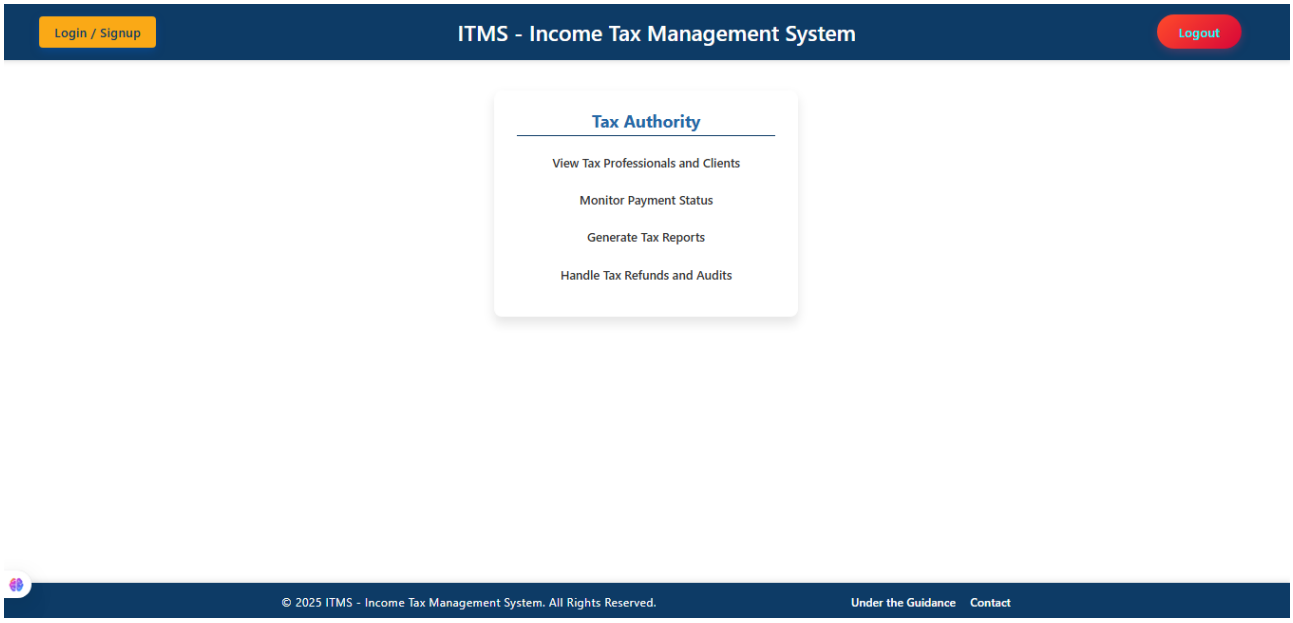

Figure 5.2.5:TaxAuthority Dashboard

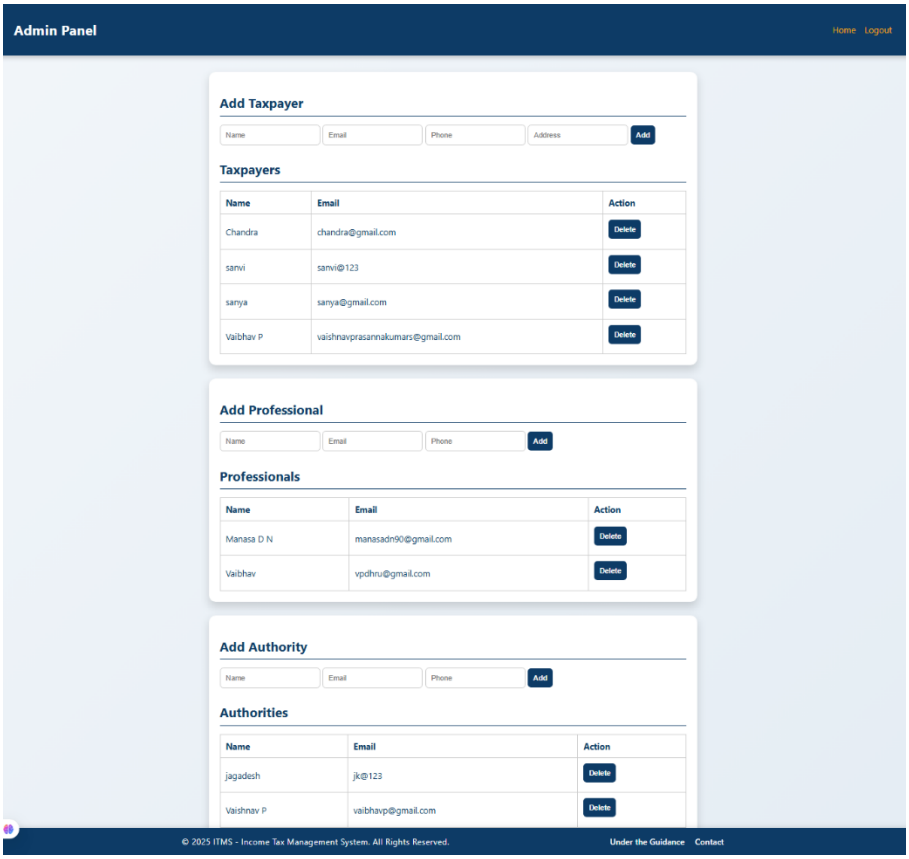

Figure 5.2.6:Admin Dashboard

5.3 Python-Based Tax Automation

Python scripts are integrated to handle **tax computation automation** based on the **New Tax Regime (FY 2023–24 onwards)**.

Automation Flow:

User Input → PHP Backend → Python Script → Output → Dashboard

Integration Workflow

1. **Frontend** — User enters income and deduction details.
2. **Backend (PHP)** — Sends data to the Python script.
3. **Python** — Calculates tax according to slab rates.
4. **Result** — Displayed dynamically in the taxpayer dashboard.

Automation Logic

```
def calculate_tax(income, deductions, prev_tax):
    taxable_income = max(0, income - deductions)
    tax = 0
    if taxable_income <= 300000:
        tax = 0
```

```
    elif taxable_income <= 600000:
        tax = (taxable_income - 300000) * 0.05
    elif taxable_income <= 900000:
        tax = 15000 + (taxable_income - 600000) * 0.10
    elif taxable_income <= 1200000:
        tax = 45000 + (taxable_income - 900000) * 0.15
    elif taxable_income <= 1500000:
        tax = 90000 + (taxable_income - 1200000) * 0.20
    else:
        tax = 150000 + (taxable_income - 1500000) * 0.30

    cess = tax * 0.04
    total_tax = tax + cess
    tax_due = max(0, total_tax - prev_tax)
    return total_tax, tax_due
```

5.4 Selenium Automation for System Testing

To validate the system's end-to-end functionality, a **Python Selenium automation script** was developed.

Automation Tools Used

- **Selenium WebDriver (Python)**
- **Google Chrome Driver**
- **Time and OS Libraries** for interaction and delays

Sample Automation Code Snippet

```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import Select
import time, os
driver = webdriver.Chrome()
signup_url = "http://localhost/income/signup.php"
```

```
login_url = "http://localhost/income/login.php"
add_revenue_url = "http://localhost/income/add_revenue.php"
def test_signup():
    driver.get(signup_url)
    driver.find_element(By.NAME, "name").send_keys("testuser")
    driver.find_element(By.NAME, "email").send_keys("testuser@example.com")
    driver.find_element(By.NAME, "phone").send_keys("9876543210")
    driver.find_element(By.NAME, "password").send_keys("TestPass123")
    Select(driver.find_element(By.NAME, "role")).select_by_value("taxpayer")
    driver.find_element(By.TAG_NAME, "button").click()
    time.sleep(2)
    assert "login.php" in driver.current_url
def test_login():
    driver.get(login_url)
    driver.find_element(By.NAME, "name").send_keys("testuser")
    driver.find_element(By.NAME, "password").send_keys("TestPass123")
    driver.find_element(By.TAG_NAME, "button").click()
    assert "taxpayer.php" in driver.current_url
test_signup()
test_login()
driver.quit()
```

5.5 Security & Validation Features

- **Session-Based Authentication** using PHP.
- **Password Hashing** before database storage.
- **Prepared Statements** to prevent SQL injection.
- **Role-Based Access Control (RBAC)** for each module.
- **File Type & Size Validation** before upload.

Chapter 6: System Testing

6.1 Overview of Testing

The **Income Tax Management System (ITMS)** underwent rigorous testing to ensure that all modules function correctly, data is processed securely, and user workflows operate without failure. The testing process focused on **functional testing, integration testing, system testing, and automated testing** using **Python Selenium WebDriver**.

6.2 Testing Techniques Used

1. **Unit Testing:**
   Each PHP module (signup, login, upload, revenue, etc.) was tested independently for functionality.
2. **Integration Testing:**
   Ensured smooth data flow between connected modules (e.g., taxpayer → professional → database).
3. **System Testing:**
   Verified complete workflow execution in a real environment using local deployment (XAMPP server).
4. **Automation Testing (Selenium):**
   Used Selenium WebDriver to automatically test form submissions, redirections, and result validations.
5. **Validation Testing:**
   Checked whether system outputs (e.g., success messages, redirects) matched expected outputs for all user interactions.

6.3 Test Cases

Table 6.1 – Functional Test Cases

Test ID	Test Case Description	Input / Action	Expected Output	Actual Result	Status
TC01	User Signup Validation	Enter name, email, password, phone, role=Taxpayer	Redirect to `login.php`	Redirect successful	✅ Passed
TC02	User Login Validation	Enter valid username & password	Open `taxpayer.php` dashboard	Dashboard loaded	✅ Passed
TC03	Invalid Login Attempt	Enter wrong password	Show "Invalid Credentials"	Error displayed	✅ Passed
TC04	File Upload Module	Select PDF file & click Upload	Message "Document uploaded successfully!"	Message shown	✅ Passed
TC05	Revenue Addition	Enter income amount ₹1000	Redirect to `taxpayer.php` with confirmation	Successful redirect	✅ Passed

Test ID	Test Case Description	Input / Action	Expected Output	Actual Result	Status
TC06	Tax Calculation (Automation)	Enter annual income ₹8,00,000	Show estimated tax per new regime	Correct tax displayed	✅ Passed
TC07	Notification Display	Professional sends comment	Taxpayer receives notification	Notification visible	✅ Passed
TC08	Admin Management	Add new user & assign role	User record saved in DB	Record inserted	✅ Passed

6.4 Result Validation

The test results confirm that:

- **Functional modules** (Signup, Login, File Upload, Tax Calculator, Notification) work as intended.
- **Database consistency** is maintained — no redundant or orphan data.
- **Selenium automation** validates real-world workflow performance.
- **Security checks** such as session validation, SQL injection prevention, and file validation are effective

6.5 Summary of Testing

The overall system testing of **Income Tax Management System** demonstrates that:

- The application is stable, reliable, and fully functional.
- Automation tests ensure consistent validation across multiple scenarios.
- The user experience remains smooth, with minimal error handling required.
- The system successfully meets the design objectives and is ready for deployment.

Chapter 7: Results and Discussion

7.1 Results

After successful implementation and testing, the **Income Tax Management System (ITMS)** produced results that meet all project objectives.The web application was deployed locally using **XAMPP** and tested with multiple user roles — **Admin, Taxpayer, Tax Professional, and Tax Authority**.All modules performed as expected, and the system functioned seamlessly with database connectivity and automation logic.

7.2 Key Functional Results

Module Description Result

User Registration and Login	Allows new users to sign up and existing users to log in securely.	Successfully verified with role-based access.
Dashboard Module	Displays role-specific options and notifications.	User-specific dashboard loaded correctly for all roles.
Document Upload	Taxpayers can upload income-related PDF documents.	File successfully uploaded and stored in /uploads directory.
Revenue Entry	Allows taxpayers to input their income for tax computation.	Revenue successfully added to database and displayed on dashboard.
Admin Control Panel	Allows admin to manage all users and monitor activity.	CRUD operations and monitoring functions working correctly.
Tax Authority Module	Provides visibility into overall taxpayer records.	Reports and verifications successfully generated.

7.4 Discussion

The **Income Tax Management System** successfully achieves its purpose of automating and simplifying the process of managing income tax activities.
Through this implementation, several observations were made:

1. **Automation Enhanced Efficiency:**
   The Python and Selenium scripts significantly reduced manual testing and computation time.
2. **Improved User Experience:**
   The responsive web interface, clear navigation, and real-time notifications ensured that users could easily interact with the system.
3. **Data Accuracy and Integrity:**
   The relational schema with foreign key constraints ensured consistent and accurate data handling between different modules.
4. **Security Strength:**
   Session management, password encryption, and prepared statements effectively protected the application from unauthorized access and SQL injection attacks.
5. **Scalability:**
   The system's modular structure allows easy addition of new roles or functionalities in the future, such as automatic tax filing or integration with official APIs.
6. **Cross-role Synchronization:**
   Communication between taxpayer and professional through notifications streamlined collaboration and improved tax compliance tracking.

Applications

The **Income Tax Management System (ITMS)** has a wide range of real-world applications, particularly in digital finance, tax consulting, and government compliance. The system supports multiple user roles and enables secure, transparent, and efficient tax management.

1. Individual Taxpayers

- Easily calculate their tax liability using automated tools.
- Upload financial documents securely.
- Receive real-time notifications from tax professionals.
- Track their yearly income, deductions, and tax payments.

2. Tax Professionals

- Manage multiple taxpayer profiles through a single dashboard.
- Review uploaded financial documents.
- Communicate with taxpayers using an integrated notification module.
- Provide timely guidance and reduce manual overhead.

3. Government/Tax Authority

- Verify taxpayer submissions and uploaded documents.
- Monitor tax records and ensure compliance.
- Generate reports for audits and policy-making.

4. Corporate Payroll & HR Departments

- Use the system to assist employees with tax documentation.
- Validate Form-16, salary slips, and declarations.
- Provide faster tax-related support.

5. Academic & Training Institutions

- As a model project to teach students about:
    - Full-stack development
    - Automation (Python + Selenium)
    - Tax computation logic and database design
    - Real-world role-based access control systems

Conclusion

The **Income Tax Management System (ITMS)** successfully integrates modern web technologies with automation to create a practical and reliable platform for managing income tax operations. Using **PHP, HTML, CSS, JavaScript, MySQL**, and **Python automation**, the system offers an end-to-end solution for tax computation, secure document handling, revenue tracking, and professional communication.

The project achieved all its objectives by:

- Providing a user-friendly interface for taxpayers and professionals.
- Ensuring accurate tax computation using the **new Indian tax regime**.
- Automating critical workflows using Python and Selenium.
- Maintaining data integrity with a well-designed relational schema.
- Ensuring secure role-based access and robust backend processing.

Overall, the system demonstrates how technology can simplify complex financial processes and support smooth and transparent tax management for individuals, professionals, and authorities.

Future Enhancements

Although the system is functional and reliable, it can be further expanded with advanced features to increase usability, automation, and real-world applicability.

1. Integration with Official Government APIs

- Auto-fetch Form-26AS, AIS, and TDS details.
- Validate PAN, Aadhaar, and tax returns directly from government servers.

2. AI-Based Document Analysis

- Automatically extract values from uploaded PDFs (Form-16, salary slips).

- Detect missing or mismatched financial data using OCR + ML.

3. Chatbot Assistance

- Integrate an AI chatbot for instant help regarding:
  - Tax queries
  - Filing instructions
  - Document support

4. Automated E-Filing

- Submit income tax returns directly from the system.
- Generate pre-filled ITR forms for taxpayers.

5. Mobile Application

- Android/iOS app for easier access for farmers, employees, and freelancers.
- Push notifications for deadlines and tax updates.

6. Multi-language Support

- Support regional languages for wider accessibility across India.

7. Advanced Analytics for Tax Professionals

- Provide insights on taxpayer patterns, pending cases, and compliance analytics.

References

Government of India, Income Tax Department. (2024). Income Tax Rules and Regulations New Tax Regime. Retrieved from https://www.incometax.gov.in/

Central Board of Direct Taxes (CBDT). (2024). Income Tax Slab Rates and Filing Guidelines. Ministry of Finance, Government of India.

Mozilla Developer Network (MDN). (2024). HTML, CSS, and JavaScript Web Documentation. Retrieved from https://developer.mozilla.org/

PHP Documentation Team. (2024). PHP Manual Server-Side Scripting Language. Retrieved from https://www.php.net/docs.php

MySQL Developers. (2024). MySQL Reference Manual Relational Database Management System. Retrieved from https://dev.mysql.com/doc/

Selenium Software Foundation. (2024). Selenium WebDriver Documentation Automated Browser Testing. Retrieved from https://www.selenium.dev/documentation/

Python Software Foundation. (2024). Python 3.10 Documentation Programming Language. Retrieved from https://docs.python.org/3/

Bootstrap Team. (2024). Frontend Components and UI Design Resources. Retrieved from https://getbootstrap.com/

Stack Overflow Community. (2023–2024). Developer Discussions and Solutions on PHP, MySQL, JavaScript, and Selenium. Retrieved from https://stackoverflow.com/

National Informatics Centre (NIC). (2023). Digital Tax Infrastructure and e-Governance Standards in India. Government of India.